LOVESSORROW

Arman Ghadimi

Software Engineering · Game Development · Unreal Engine

Dissonant and Desolate

PORTFOLIO — 2026

01 / PROFILE

About Me

I am a Computer Engineering student at Sharif University of Technology, focused on game development, software systems, Unreal Engine tooling, and AI-assisted engineering. I build modular tools, gameplay systems, and technical prototypes across Unreal Engine, Unity, LibGDX, and Minecraft, with a particular interest in plugin architecture, rendering research, game systems, and interactive storytelling.

Computer Engineering — Sharif University of Technology
Game Development · Software Engineering · Creative Technology

github.com/LovesSorrow

FOCUS

Unreal Engine & C++
Plugin development, editor tooling, gameplay systems

Game Systems & Architecture
Modular systems, ECS, extensible gameplay design

AI-Assisted Development
Planning, implementation, review, and technical iteration

Rendering & Technical Art Research
Anime-style rendering, materials, environments

Creative Development
Interactive storytelling, game design, literature, music

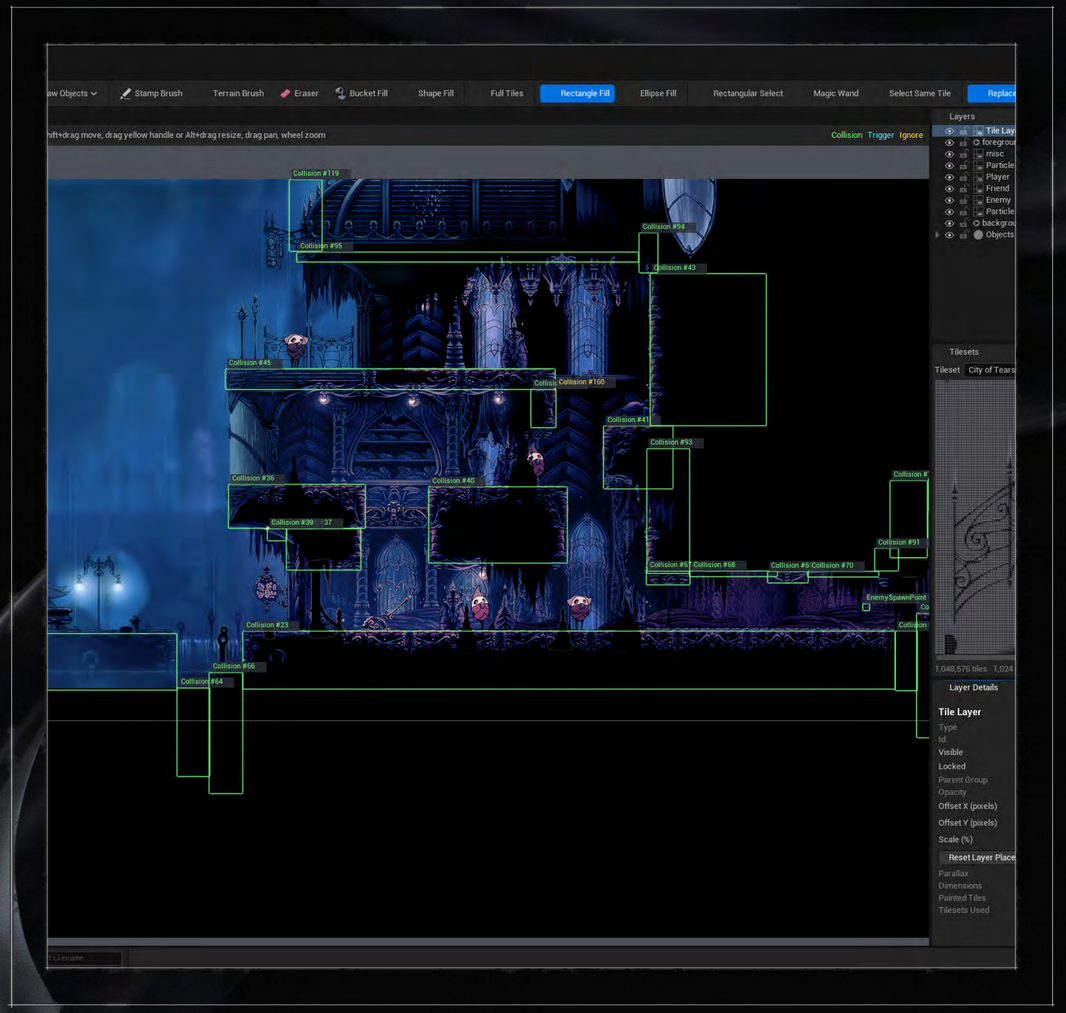
02 / FEATURED PROJECT

Tiled Importer

Unreal Engine Plugin · C++ · Editor Tooling

An in-development Unreal Engine plugin focused on bringing Tiled workflows into Unreal through modular import systems and extensive editor tooling. The project goes beyond basic map import, aiming to reproduce and extend important parts of the Tiled authoring experience directly inside Unreal Engine.

Status	In Development
Engine	Unreal Engine
Focus	Plugin Architecture · Editor Tooling · Workflow Integration
Development	AI-Assisted



UNREAL EDITOR / TILED IMPORTER

From Tiled Data to Unreal Workflow

A modular pipeline connecting Tiled map data, Unreal assets, editor tooling, and scene generation.



IMPORT & PARSING

Finite & infinite maps
CSV / Base64 data
zlib / gzip compression
Tile, object & group layers
Embedded & external tilesets
Properties & GID transformations

EDITOR WORKFLOW

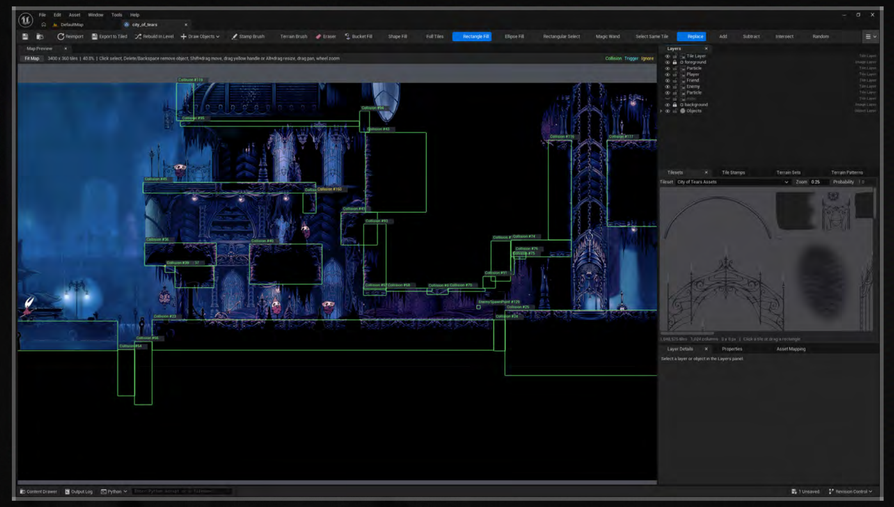
Import & reimport
Layer visibility / locking / ordering
Layer grouping & duplication
Map preview
Property editing
Object manipulation

SCENE GENERATION

Paper2D tiles & sprites
Layered scene construction
Collision generation
Triggers
Class-based actor spawning
Editor preview

Recreating the Workflow, Not Just the File Format

The plugin extends beyond import into editor-side workflows for inspecting, organizing, and modifying Tiled content inside Unreal Engine.

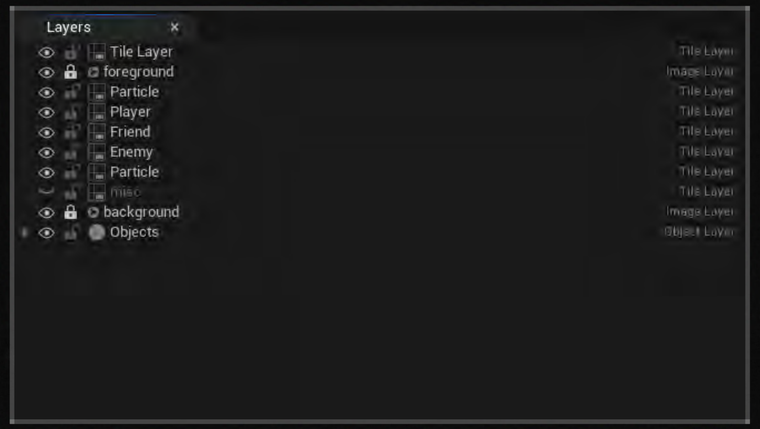


MAP PREVIEW / LAYER WORKFLOW

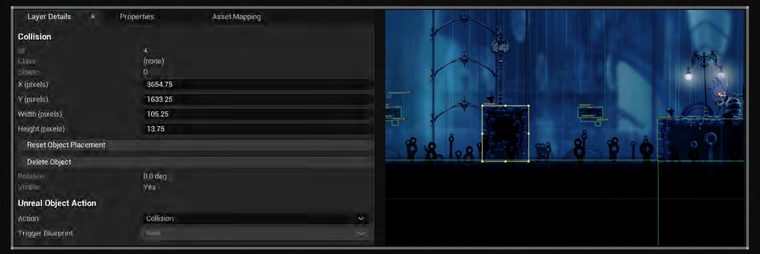
The editor tooling is designed to keep imported Tiled content inspectable and editable inside Unreal, reducing the gap between external map authoring and Unreal-side iteration.

SELECTED EDITOR CAPABILITIES

- Layer visibility, locking, ordering & grouping
- Duplicate, delete & multi-selection workflows
- Map and layer preview
- Property editing across maps, layers and objects
- Object move / resize / shape editing
- Undo-supported editor operations



LAYER MANAGEMENT



PROPERTIES / OBJECT EDITING

05 / PROTOTYPE

Hollow Knight — Unreal Prototype

A gameplay and level prototype built in Unreal Engine to exercise the Tiled Importer in a real project context.

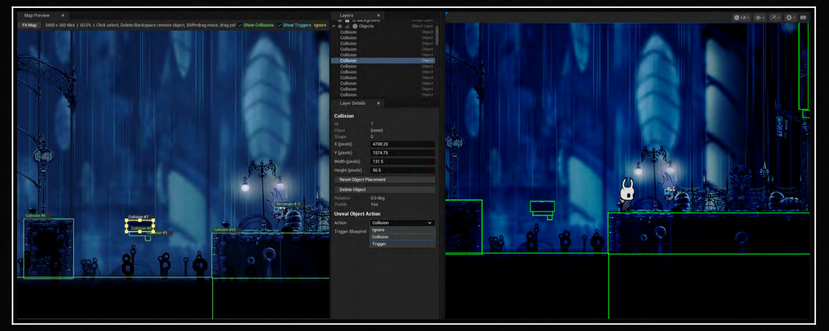
This in-development prototype uses a Hollow Knight-inspired project context to test the Tiled Importer beyond isolated editor demonstrations. It provides a practical environment for validating imported maps, scene construction, collisions, object placement, and the broader Unreal-side workflow.



UNREAL PROTOTYPE / IMPORTED LEVEL



COLLISION / SCENE VALIDATION



OBJECT / LAYER INTEGRATION

WHAT THIS PROTOTYPE VALIDATES

- Imported map and scene representation
- Collision generation and placement
- Layered level construction
- Object and actor placement
- Reimport and iteration workflow
- Using the plugin within an actual game prototype

Status	In Development
Engine	Unreal Engine
Role	Technical Prototype / Tiled Importer Showcase
Focus	Level Integration · Tool Validation · Gameplay Prototyping

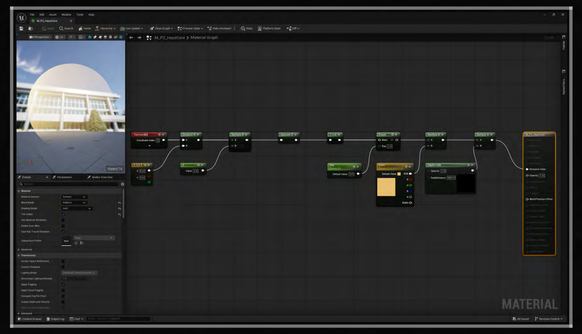
Wandering Witch

Unreal Engine Vertical Slice · Rendering Research · RPG & Narrative Design

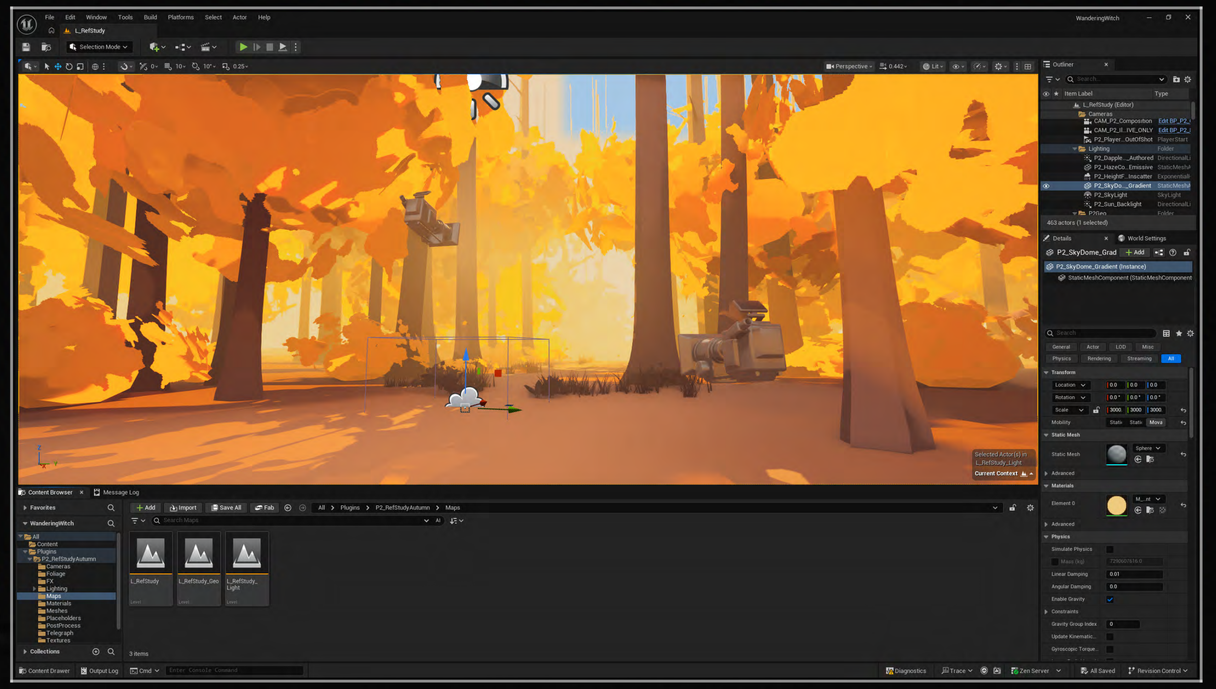
A long-term Unreal Engine project exploring the kind of game I ultimately want to create. Wandering Witch serves as a vertical slice for exploring anime-style rendering, mage-oriented gameplay, RPG systems, environments, 3D production, and interactive storytelling while developing the technical and creative knowledge required for larger future projects.



CHARACTER / APPEARANCE RESEARCH



RENDERING / MATERIAL RESEARCH



VISUAL DEVELOPMENT / UNREAL ENGINE

PROJECT RESEARCH

RENDERING
Anime-style rendering, materials, lighting, and visual presentation.

GAME DESIGN
RPG structure, exploration, progression, and systemic gameplay.

MAGE MECHANICS
Wind, fire, telekinesis, defensive abilities, movement, and ability interaction.

WORLD PRODUCTION
3D modeling, environments, scene composition, and technical-art workflows.

NARRATIVE
Interactive storytelling, adaptation, original writing, and episodic narrative structure.

Status	Research & Design
Engine	Unreal Engine
Type	Vertical Slice / R&D Project
Development	AI-Assisted
Current Phase	Rendering Research · Game Design · Technical Planning

The project combines technical research with creative development: rendering shapes the visual language, gameplay systems support the fantasy of being a traveling mage, and narrative gives those systems context.

Personal research project inspired by
Wandering Witch: The Journey of Elaina.

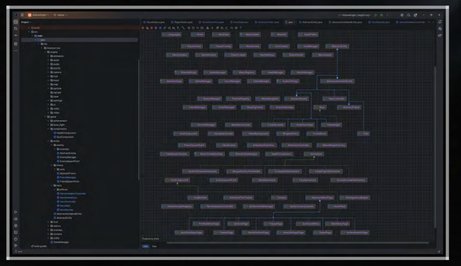
Hollow Knight — LibGDX Prototype

Java · LibGDX · Modular Architecture · Tiled Integration

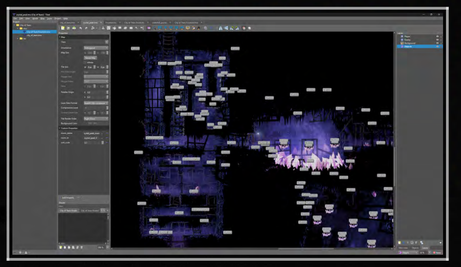
Developed as part of university coursework, this prototype recreates selected 2D action-platformer systems using Java and LibGDX. The project emphasizes modular architecture, Tiled-based level integration, reusable gameplay systems, and implementation closer to the framework level rather than relying on a full game engine.



GAMEPLAY



INHERITANCE IMPLEMENTATION / PROJECT STRUCTURE



TILED MAP / LEVEL PIPELINE

SYSTEM DESIGN

Tiled Level Data → World / Level Representation → Gameplay Systems → Rendering & Animation

Type	University Project
Framework	LibGDX
Language	Java
Focus	Game Systems · Modular Architecture · Tiled Integration

LEVEL SYSTEM
Tiled-based map loading and level organization

MOVEMENT & PHYSICS
2D platforming, collision handling, and player movement

COMBAT
Reusable combat and gameplay-state logic

ANIMATION
Animation state handling and gameplay-driven transitions

ARCHITECTURE
Modular systems designed to remain separated and extensible

Unlike the earlier Unreal prototype, this project emphasizes framework-level game-system implementation in Java and LibGDX.

Unofficial university prototype inspired by Hollow Knight; not affiliated with Team Cherry.

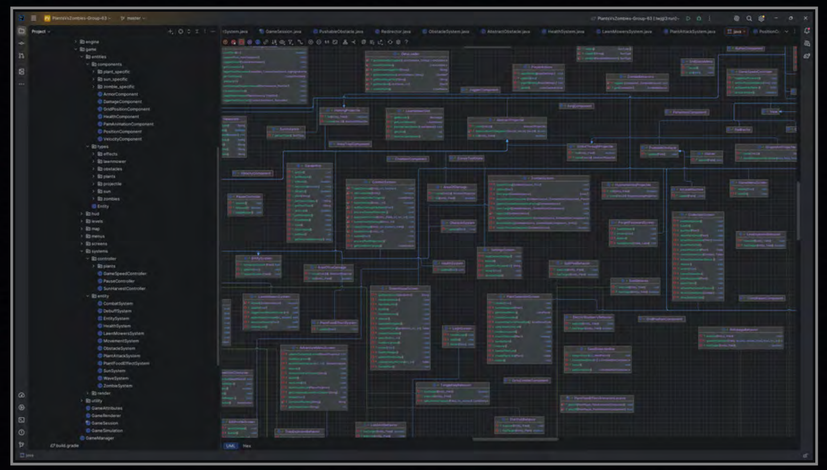
Plants vs. Zombies — LibGDX Prototype

Java · LibGDX · Entity Component System · Team Project

Developed as a university team project, this Plants vs. Zombies recreation uses a modular Entity Component System architecture to separate game data from behavior. The structure is designed to support many plants, zombies, and gameplay behaviors without tightly coupling every entity type to its implementation.



GAMEPLAY / SYSTEM INTEGRATION



ECS IMPLEMENTATION / PROJECT STRUCTURE

ECS STRUCTURE

ENTITIES
Plant · Zombie
Projectile · Resource

→

COMPONENTS
Position · Health · Attack
Movement · Targeting · Production

→

SYSTEMS
Movement System · Combat System
Targeting System · Resource System

Entities are composed from data components; systems operate on entities with matching component sets.

Type	University Team Project
Framework	LibGDX
Language	Java
Architecture	Entity Component System
Focus	Extensible Gameplay · Modular Systems · Team Development

DESIGN GOALS

MODULARITY
Separate gameplay data from system behavior

EXTENSIBILITY
Add new entity combinations without rebuilding the entire architecture

REUSABILITY
Shared systems operate across compatible entities

SCALABILITY
Architecture designed to support many plants, zombies, and behaviors

TEAM DEVELOPMENT
Clearer system boundaries support collaborative implementation

Unofficial university recreation inspired by Plants vs. Zombies; not affiliated with PopCap or Electronic Arts.

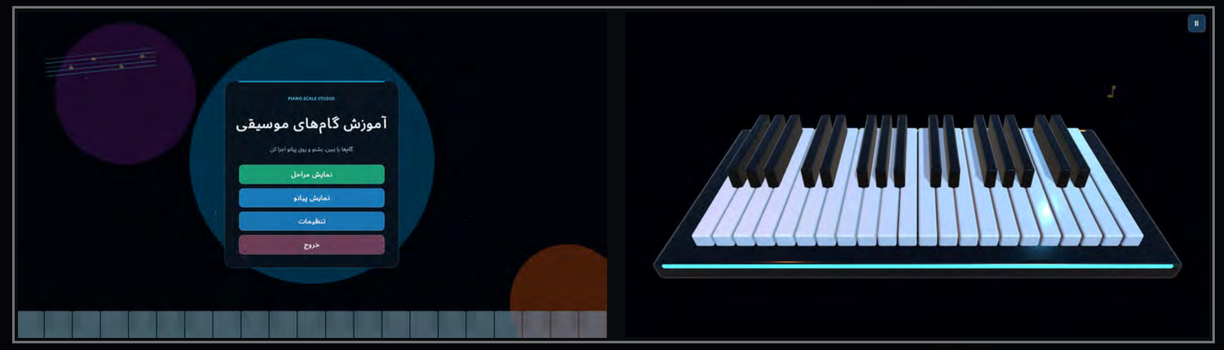
Unity Projects

Freelance Delivery · Gameplay Prototyping · AI-Assisted Development

These Unity projects represent two different forms of applied development: a delivered freelance educational tool and a gameplay recreation used as a foundation for studying game design, implementation, and later creative iteration. Both projects were developed through an AI-assisted programming workflow with manual review, debugging, and integration.

MusicScaleTrainer

Unity · Freelance Project · Educational Interaction



INTERACTIVE PIANO / SCALE TRAINING

A freelance Unity project built around an interactive piano that allows the user to play notes and practice musical scales. The project combines direct musical interaction with a focused learning experience and was delivered as a working application.

PROJECT FOCUS

- Interactive piano input
- Musical scale learning workflow
- Audio / interaction feedback
- Unity implementation
- Freelance delivery

Status	Delivered
Engine	Unity
Type	Freelance Project

Bridge Race Recreation

Unity · Game Design Learning Project



GAMEPLAY / RECREATION BASELINE

A Unity recreation used as part of a game-design learning path. The current implementation establishes the recognizable gameplay baseline, while later iterations are intended to introduce original creative and design decisions rather than remain only a recreation.

LEARNING FOCUS

- Gameplay-loop reconstruction
- Mechanic analysis
- Implementation practice
- Game-design iteration
- Planned original design iterations

Status	Ongoing
Engine	Unity
Type	Game Design / Learning Project

AI-ASSISTED DEVELOPMENT WORKFLOW

AI supports implementation and iteration, while project direction, review, debugging, integration, validation, and final technical decisions remain part of my development process.

Requirements → Implementation → Review → Debugging → Iteration

Writing Beyond the Game

Short Friction · Narrative Development · Original Piano Composition

My interest in game development extends into independent writing and music. These works provide a space to explore characters, atmosphere, narrative structure, and emotion outside interactive systems, while also developing creative skills that can later inform storytelling inside games.

A Cruel Sign Drifting In The Dark

نشانه ای ظالم پرسه زنان در تاریکی

Three-Episode Short Story — In Development

An in-development three-episode short story exploring the history and characters connected to a larger narrative world. The project is currently in its creative-design and writing phase.

ORIGINAL MUSIC

Piano Composition

What Remained of That Night

An original piano piece created in connection with the story, extending its emotional and atmospheric direction through music.



STORY / VISUAL DEVELOPMENT

CREATIVE PRACTICE

NARRATIVE
Character, world, and story development

ATMOSPHERE
Mood, emotional direction, and thematic presentation

MUSIC
Using piano composition to extend narrative emotion

GAME DEVELOPMENT
Applying narrative and creative experience to future interactive projects

Format	Three-Episode Short Story
Status	In Development
Focus	Writing · Narrative Design · Character & World Development
Music	Original Piano Composition

Writing and music are separate creative practices, but both strengthen the narrative and emotional skills I want to bring into long-term game development.

Engineering, Games & Creative Systems

A developing technical practice across tools, gameplay, rendering, software architecture, and interactive storytelling.

My work spans game-engine tooling, gameplay systems, framework-level implementation, rendering research, applied development, and narrative experimentation. Across these projects, I aim to understand not only how systems are built, but how technical decisions, player experience, visual direction, and creative intent connect inside larger interactive projects.

CORE DEVELOPMENT

LANGUAGES

C++ **Java** **C#**

ENGINES & FRAMEWORKS

Unreal Engine

Unity

LibGDX

ENGINEERING AREAS

Unreal Engine plugin & editor tooling

Gameplay systems

Modular software architecture

Entity Component Systems

Tiled map integration

Rendering & technical-art research

CREATIVE AREAS

Game design

Interactive storytelling

Narrative development

Anime-style rendering research

Environment development

Piano composition

EDUCATION

B.Sc. Computer Engineering
Sharif University of Technology
In Progress

CONTACT

GitHub **github.com/LovesSorrow**
Email **armanghadimi86@gmail.com**
Discord **snowsorrow**

SELECTED TOOLS

Git · Blender · Visual Studio / Rider · Claude / Codex

LovesSorrow

Dissonant and Desolate